Problem Set 3 Solution
ECO3121 - Fall 2023

November 26, 2023

In this problem set, we'll continue to examine the effects of land rental activity in rural households on output and aggregate productivity. Please continue using the main dataset `"aghousehold.dta"` from your first assignment from the blackboard site. It is the National Fixed Point Survey (NPFS) in the year of 2010.

We continue with our inquiry on the causal effect of land rental behavior on agricultural productivity. The researcher plans to follow the specifications in Question 1-7 in Assignment 1 and regress the yield on land_rental-out activities..

Model 1:

$$logyield_i = \beta_0 + \beta_1 rental_out_share_i + \mu_i$$

Question 1

1. List three plausible arguments why the point estimate in the regression model 1 could be biased, and the corresponding bias directions (**upwards** or **downwards**) relative to the true causal effect of household's rental activities on household yield. (6 points)

   [Open Question] Answers:

   1. Omitted variable bias - positive bias. For example, land management services. Precise land management services could typically increase the yield upon the land, and more rental out activity can grant limited land management services to be applied upon more compact plots of land, resulting in more precise provision of services and therefore higher yields.

   2. Omitted variable bias - negative bias. For example, household's characteristics. A lazy household is more likely (free) to rent out the land and is less likely to have a higher yields.

   3. Measurement errors in explanatory variable rental_out_share - negative bias. A measure error can always cause attenuation bias (the estimation is more closer to zero than the true parameter), and since we have already known the effect of rental out activity on yields is positive, then we can infer the bias is negative.

2. Now your friend suggests that the total land area ($d31$ in the data) of each household could be a valid instrumental variable (IV) for your measure of household's land rental-out share $rental_out_share_i$ (the proportion of rent-out land to total land area (d31)) . List the assumptions that need to hold true for this to be correct and explain why this instrumental variable is true (or not true). (4 points)

Answers: ***Relevance.*** *$Cov(total_land, rental_out_share) \neq 0$. Total land area should have direct effects on land rental-out in share form.*

Exogeneity. *$Cov(total_land, \mu) = 0$. Total land area should not be correlated with μ, that is not have direct effects on yields.*

[Open Question] Apparently, the Exogeneity assumption does not hold true. Because the total land area is actually correlated with μ, for example, agriculture technologies can be included in μ and larger total land is applicable with more advanced agriculture technologies.

3. Now your TA suggests that the rainfall (precipitation) could be a valid instrumental variable (IV) for your measure of household's land rental-out share $rental_out_share_i$. Try to merge the household's production dataset and precipitation dataset via *vl_id*, the specific village identifier, using stata command *merge* (Many-to-one merge, type *"help merge"* in Stata for assistance).

   Note: drop the missing value for $rental_out_share_i$ av_rain **and** $logyield_i$ **First!**

   You decide to use the average rainfall in 2010 (*av_rain*) as the IV for the household's rental-out land share. Verify if the assumption of instrument relevance is (or is not) satisfied using the first stage regression, and export the results. **Write down the first stage regression model** and interpret the result and statistical significance of your result. (5 points)

   Answers: The first stage regression is:

$$rental_out_share_i = \pi_0 + \pi_1 av_rain + \nu_i$$

   The regression coefficient is 0.00791 in column (3). It means that a 100 mm increase in precipitation will cause 0.791 unit increase in rental-out land in share. It's statistically significant at 1% level.

4. Now use Stata/R to estimate the 2nd stage IV point estimate (using linear model) as suggested by the TA, and export your result. **Write down the second stage regression model** and interpret the result and statistical significance of your result. (5 points)

   Answers: The second stage regression is:

$$log\ yield_i = \beta_0 + \beta_1 \widehat{rental_out_share_i} + \mu_i$$

   The regression coefficient is 0.00159 in column (4). It means that one unit increase in share of rental-out land will cause a 0.159% increase in yield. It's ***NOT*** *statistically significant under 1% level.*

5. Now your TA tells you that you can use *ivreg* command directly to replicate the results in question (4). Do you find any difference in the IV estimations (β_{IV}) **regarding the coefficients and standard errors** relative to (4). (3 points)

   Answers: See regression table column (5) for details. When appliying ***ivregress****, the magnitude of the coefficients are the same as 0.00159 for questions (4) and (5). But the standard errors are different (0.00143 vs. 0.00145) since the standard error in the second stage regression in (4) is incorrect due to lacking consideration of first-stage regression.*

	(3)	(4)	(5)	(6-Rel)	(6-IV)	(6-Exo)
Depend Var	rental_out_share	log yield	log yield	rental_out_share	log yield	IV-residual
av_rain	0.00791***			0.00755***		-0.00002
	(0.00131)			(0.00132)		(0.00001)
$\widehat{rental_out_share}$		0.00159				
		(0.00143)				
rental_out_share			0.00159		0.00336***	
			(0.00145)		(0.00129)	
implemented				4.78609***		0.02151*
				(1.26662)		(0.01160)
constant	1.25257	5.92412***	5.92412***	-1.68400	5.90903***	-0.00032
	(1.34547)	(0.01326)	(0.01340)	(1.55323)	(0.01225)	(0.01423)
Observations	13,897	13,897	13,897	13,897	13,897	13,897
R-squared	0.0026	0.0001	.	0.0036	.	0.0003

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Figure 1: All regression results in Assignment 3.

6. Now your Professor suggests that both rainfall (Z_1) and the implementation of land law (Z_2) which legalizes the land transaction could be valid instrumental variables (IVs) for your measure of household's land rental-out share $rental_out_share_i$. Try to merge the household's production dataset and land law dataset named "landlaw.dta". To do the merge, you should take the first two digit number of *vl_id* in the aghousehold data by using the command "substr" and then generate a new variable called *vl_id*2. Next, merge household data and landlaw data via *vl_id*2, the specific province identifier, using stata command *merge* (Many-to-one merge, type *"help merge"* in Stata for assistance).

   you can use *ivreg* command directly to get the estimations. (15 points)

   1) Propose the tests for the relevance assumption and exogenous assumption under this context. **Write down the regression model** and each step of the two tests.

   Answers: **Relevance Assumption:**

   - *Conduct following first stage regression:*

$$rental_out_share_i = \pi_0 + \pi_1 av_rain_i + \pi_2 land_law_i + \nu_i$$

   - *Choose a null hypothesis:*

$$H_0 : \ \pi_1 = \pi_2 = 0.$$

   - *Calculate F-statistic, if $F < 10$, the model might have weak IV issue.*

   Exogenous Assumption: *With $m = 2 > k = 1$, we can use J-test:*

- *Conduct following 2SLS regression:*

$$log\ yield_i = \beta_0 + \beta_1 rental_out_share_i + \mu_i$$
$$rental_out_share_i = \pi_0 + \pi_1 av_rain_i + \pi_2 land_law_i + \nu_i$$

- *Obtain the estimator* $\hat{\beta}_0^{2SLS}$, $\hat{\beta}_1^{2SLS}$, *compute the predicted value* $\widehat{log_yield}$ *using actual rental_out_share;*
- *Compute residual* $\hat{u} = \widehat{log_yield} - log_yield$*;*
- *Regress* $\hat{u}$ *on all IVs and exogenous variables, here we conduct:*

$$\hat{u}_i = \delta_0 + \delta_1 av_rain_i + \delta_2 land_law_i + \epsilon_i$$

- *Choose a null hypothesis:*

$$H_0 :\ \delta_1 = \delta_2 = 0.$$

- *Calculate F-statistic, and the J-statistic:*

$$J = 2F \sim \chi^2_{(2-1)}$$

- *Choose a certain significance level, here we choose 5%, the critical value under* $d.f. = 1$ *is 3.84. If J-test rejects, then at least some IVs are endogenous.*

2) Conduct the tests using our data.

- **Relevance Assumption:** *See regression table column (6-Rel) for details.*
  We can calculate R^2_{ur} *from STATA output, and further F-statistic:*

$$R^2_{ur} = \frac{243039.2}{66982436.1} = 0.0036284$$
$$F = \frac{R^2_{ur}}{k} \cdot \frac{n-k-1}{1-R^2_{ur}} = \frac{0.0036284 * 13894}{2 * (1-0.0036284)^2} = 25.298$$

  Since $25.298 > 10$*, the two instruments aren't weak.*

- **Exogenous Assumption:** *See regression table column (6-IV) for IV-regression details, and column (6-Exo) for J-test regression details.*
  We can calculate R^2_{ur} *from STATA output, and further F & J-statistic:*

$$R^2_{ur} = \frac{1.92907}{5599.63174} = 0.0003445$$
$$F = \frac{R^2_{ur}}{k} \cdot \frac{n-k-1}{1-R^2_{ur}} = \frac{0.0003445 * 13894}{2 * (1-0.0003445)^2} = 2.394$$
$$J = 2 * F = 4.788$$

  Since $4.788 > 3.84$*, we can reject the null hypothesis under 5% significance level, which means that either av_rain or landlaw implemented is endogenous, and even both are endogenous.*